Rahul Dey

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PROFILE

MSc Bioinformatics student at the University of Edinburgh with experience in single-cell transcriptomics, RNA sequencing analysis and reproducible computational workflow development. Proficient in R (Seurat, Bioconductor) and Python (Scanpy) for analysing large-scale transcriptomic datasets, with a strong interest in computational genomics, functional genomics and developing scalable bioinformatics software. Experienced in Linux and HPC environments, with a focus on reproducible research and software engineering.

EDUCATION

- THE UNIVERSITY *of* EDINBURGH — Edinburgh, UK
  M.Sc. Bioinformatics — (2025-2026)
- SISTER NIVEDITA UNIVERSITY — Kolkata, India
  B.Sc. (Hons.) Biotechnology — First Class — (2022-2025)

PROJECTS

Single-Cell RNA-seq Analysis of Mouse Embryonic Stem Cells

Tools: R (Seurat; Bioconductor: SingleCellExperiment, scran, scater), Python (Scanpy, AnnData), Kallisto, FastQC, MultiQC, Trimmomatic, Git

GitHub: https://github.com/rahuldey5156/Single-Cell-RNA-seq-Analysis

- Reproduced and extended a published single-cell transcriptomic study (Kolodziejczyk et al., Cell Stem Cell 2015) using the E-MTAB-2600 dataset (869 cells, 3 conditions).
- Implemented end-to-end scRNA-seq pipelines in both R (Bioconductor, Seurat) and Python (Scanpy), including QC, normalisation, HVG selection, dimensionality reduction, and clustering.
- Compared methodological differences across frameworks (scran vs Seurat vs Scanpy) and assessed their impact on clustering and gene variability.
- Identified a rare 2C-like totipotent subpopulation using Zscan4-family gene expression and correlation-based marker discovery.

Analysis of MTR4-Mediated RNA Surveillance in Mouse Spermatogenesis

Tools: R (DESeq2, fgsea), STAR, featureCounts, FastQC, MultiQC, BiomaRt, Linux, Git

GitHub: https://github.com/rahuldey5156/Functional-Genomic-Technologies

- Analysed a large-scale RNA-seq dataset (GSE253154) to investigate the role of MTR4 in RNA surveillance during spermatogonial differentiation.
- Performed end-to-end processing including quality control, genome alignment (STAR), and gene-level quantification, followed by differential expression analysis in DESeq2 with dispersion modelling and FDR correction.

- Conducted exploratory analysis (PCA, clustering) to assess sample variability and detect subtle transcriptomic differences between conditions.
- Identified a small set of differentially expressed genes and performed pathway enrichment analysis (fgsea, MSigDB), revealing global transcriptional downregulation trends.
- Developed a reproducible, version-controlled workflow using Git, ensuring transparency and reusability of the analysis.

ProtExplorer — Web-Based Protein Sequence Analysis Tool

Tools: PHP, MySQL (PDO), Python, Clustal Omega, EMBOSS patmatmotifs, BioPython, matplotlib, D3.js, Apache, Git

GitHub: https://github.com/rahuldey5156/IWD2

Link: `https://bioinfmsc8.bio.ed.ac.uk/~s2793337/Website/`

- Developed a full-stack bioinformatics web platform for automated comparative protein sequence analysis, enabling users to retrieve, align and analyse protein sequences across taxonomic groups through an integrated computational workflow.
- Implemented a six-stage analysis pipeline integrating NCBI Entrez (BioPython), Clustal Omega alignment, per-position conservation scoring, EMBOSS patmatmotifs PROSITE motif scanning, phylogenetic tree construction and AlphaFold structure lookup via REST API.
- Designed a normalised relational MySQL database schema with four tables, enforcing strict separation of concerns — all database interactions via PHP PDO prepared statements, with Python handling computation only.
- Developed an interactive phylogenetic tree visualisation using D3.js with linear and radial layouts, zoom, pan and node collapse/expand, rendered client-side from Clustal Omega Newick output.
- Engineered robust error handling including NCBI retry logic and deployed a pre-processed example dataset with multi-audience biological documentation.

Automated RNA-seq Analysis Pipeline Using Bash on HPC Systems

Tools: Bash, FastQC, Bowtie2, Bedtools, Linux, HPC (SLURM), Git

GitHub: https://github.com/rahuldey5156/My-FIrst-Pipeline

- Developed automated RNA-seq processing workflows using Bash scripting in a Linux environment.
- Integrated tools including FastQC, Bowtie2 and Bedtools to perform quality control, alignment and downstream data processing and used Git for version control.
- Executed RNA-seq analysis pipelines on HPC clusters using SLURM for job scheduling, resource allocation, and debugging workflow failures.

GitHub: https://github.com/rahuldey5156/Bioinformatics-Programming-And-Systems-Management

- Built a modular Python program pipeline for protein sequence profiling with automated taxonomic filtering and reproducible output generation.
- Designed and implemented the program using variables, loops, and error traps, and wrote a clear user manual explaining execution, outputs, and interpretation for non-specialist users.
- Used Git/GitHub for version control, collaborative development and pipeline reproducibility.
- Planned, structured, and prioritised development tasks to deliver the full workflow and documentation within coursework deadlines.

RESEARCH EXPERIENCE

MSc Dissertation (Ongoing)
The Roslin Institute, The University of Edinburgh
Title: Developing an R package for model-based k-mer analysis
Supervisor: Dr. Hannes Becher

Tools: R, R package development (devtools, roxygen2), Shiny, Git
GitHub: https://github.com/rahuldey5156/shiny-k-mers/tree/CRAN

- Extending and refactoring the existing Tetmer R package for CRAN submission, improving usability, documentation, and code structure.
- Implementing additional functionality, including estimation of PCR duplication levels and enhanced genome profiling metrics.
- Developing unit tests to ensure robustness and reproducibility of statistical modelling functions.
- Designing a Shiny interface to facilitate user-friendly interaction with k-mer–based genome profiling workflows.

Undergraduate Research Trainee **March 2023 - June 2025**
Department of Biotechnology, Sister Nivedita University
Supervisor: Dr. Asis Jana

Tools: GROMACS, Python, TCL scripting, Linux, HPC

- Conducted atomistic MD simulations in GROMACS to analyse stability and dynamics of tau filaments across tauopathies, and computed end-state free energies to identify determinants of differential stability.
- Wrote TCL scripts to analyse simulation trajectories, and employed Python libraries for data analysis and visualisation to extract insights from large datasets.
- Rendered structural representations of biomolecules using Blender, iteratively refining models to reveal mechanisms underlying structural polymorphism in tau fibrils.

Summer Research Intern **May 2024 - July 2024**
Department of Biological Sciences, IISER Kolkata
Supervisor: Dr. Arnab Gupta

Tools: GROMACS, Python, Linux, HPC, NumPy, Matplotlib, PyTorch/TensorFlow

- Explored the differential binding of the Kinesin Tail Domain (KTD) to monolayer and bilayer lipid vesicles, identifying a DOPA binding region in the protein as a potential therapeutic target.
- Conducted molecular dynamics simulations in GROMACS to study the behavior of KTD in different environments, performing large-scale simulations on high-performance computing (HPC) clusters for efficient computation.
- Leveraged Python programming and Linux systems to process, visualise and interpret large datasets, and implemented reproducible data analysis workflows.
- Trained an autoencoder model to generate the latent space of proteins in monolayer and bilayer environments.
- Analysed complex data sets and collaborated with the research team to design and execute research plans, and to discuss and interpret results.

SKILLS

- **Programming & Scripting:** R, Python, Bash
- **Bioinformatics Tools:** Seurat, Scanpy, Kallisto, tximport, scran, scater, SingleCellExperiment, FastQC, MultiQC, STAR, HISAT2, Bowtie2, Bedtools, featureCounts
- **Bioinformatics & Omics Analysis:** Single-Cell RNA-seq analysis (QC, normalisation, dimensionality reduction, cell type characterisation), RNA-seq analysis (QC, alignment, quantification, differential expression), transcriptomics, functional genomics, pathway enrichment (MSigDB, fgsea)
- **Workflow Development & Reproducibility:** Snakemake pipeline development, Git/GitHub version control, reproducible research practices
- **Computing Environments:** Linux/Unix systems, high-performance computing (HPC) using SLURM

- **Data Analysis & Statistics:** Statistical modelling, exploratory data analysis (PCA, clustering), data visualisation
- **Software Engineering & Web Development:** R package development, unit testing, Shiny application development, Full-stack development (LAMP stack), RESTful API design & integration, Backend pipeline development (Python, Bash),
- **Databases & Data Systems:** Relational databases (MySQL, PostgreSQL, SQL schema design), Data modelling and normalisation

PUBLICATIONS

1. **Rahul Dey** and Asis K. Jana*, The Stability and Dynamics of Tau Filaments (Manuscript under preparation).
2. **Rahul Dey**, Siddhartha Dutta and Asis K. Jana*, Understanding the Molecular Mechanism of Calmodulin7- DNA Interaction in *Arabidopsis thaliana* (Manuscript under preparation).

AWARDS & SCHOLARSHIPS

School of Biological Sciences South and South-East Asia Academic Excellence Masters Scholarship
The University of Edinburgh | Awarded: 2025
Value: £10,000 | For MSc in Bioinformatics (2025–2026)